RSG und FFGFT: Ein struktureller Vergleich

Recursive Survival Geometry und
Fundamentale Fraktale Geometrische Feldtheorie
Komplementarität, Asymmetrie und die Frage der Vollständigkeit

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Inhaltsverzeichnis

Abstract

Recursive Survival Geometry (RSG, Austin 2024–2026) and Fundamental Fractal Geometric Field Theory (FFGFT/T0, Pascher 2024–2026) address overlapping questions — what physical structure persists, and why — from structurally different starting points. This document compares the two frameworks along four dimensions: what each framework takes as primitive, what it derives, where the two accounts are complementary, and where FFGFT goes deeper in the sense of producing numerical predictions that RSG does not. The conclusion is that the two frameworks are asymmetrically related: FFGFT provides the scale structure that RSG requires but does not derive; RSG provides a general selection principle that FFGFT does not formulate explicitly. Neither is an overlay or a bridge layer of the other.

.1 What RSG Is

Recursive Survival Geometry is Peter Austin's framework for observable physical structure as the differential persistence of histories under a recursive survival-weighted filter. The central equation is:

$$\frac{dS_i}{dt} = -\Gamma_i \cdot W_i \cdot S_i \tag{1}$$

where S_i is the survival weight of history i, Γ_i is the loss rate, and W_i is the survival weight. What we observe as matter, radiation, or structure are the histories that persist under this filter. Histories that fail to persist below the threshold are not observed.

RSG's ontological primitive is the *history* — a path through configuration space — and the *survival filter* — the operator that weights persistence. The filter is general: RSG does not commit to a specific Lagrangian, a specific field content, or a specific set of coupling constants. It describes the structural logic of selection without specifying what is being selected from.

What RSG derives

RSG derives the structural conditions under which a history can persist: it must survive repeated application of the filter, it must maintain closure under the recursive process, and it must accumulate weight W_i above threshold. The formalism produces a general account of why some configurations persist and others do not.

What RSG does not derive

RSG does not derive the values of physical constants. The loss rates Γ_i, the survival weights W_i, and the threshold conditions are framework parameters. RSG does not explain why the electron mass is 0.511 MeV, why the fine structure constant is $\alpha \approx 1/137$, or why the CMB temperature is 2.725 K. These are inputs to RSG's filter, not outputs of it.

.2 What FFGFT Is

Fundamental Fractal Geometric Field Theory (T0/Time-Mass Duality) takes the 4-dimensional torus $T^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$ with winding structure $(n_\theta, n_\phi, n_3, n_4)$ as its ontological primitive. The foundational relation is $\tilde{T} \cdot m = 1$ (time-mass duality). A single dimensionless parameter $\xi = 4/30000$ is determined from three measured mass ratios via the unique winding exponent $\kappa = 7$.

The T0 Lagrangian is:

$$\begin{aligned}\mathscr{L}_{\text{T0}} = &-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell \\ &+ \tfrac{1}{2}(\partial_\mu \Delta m)(\partial^\mu \Delta m) - \tfrac{1}{2}m_T^2 \Delta m^2 + \xi \cdot m_\ell \cdot \bar{\psi}_\ell \psi_\ell \cdot \Delta m\end{aligned} \tag{2}$$

where $\Delta m(x,t)$ is the dynamical mass field and $\xi \cdot m_\ell$ is the vertex coupling.

What FFGFT derives

From $\xi = 4/30000$ alone:

- Lepton and quark mass ratios via $\kappa = 7$ recursion (error $< 1\%$, no fitting)
- Fine structure constant α (Doc 009/137)
- CMB temperature T_{CMB} (deviation $\Delta = 0.0002\%$, Doc 025)
- Cosmological constant $\Lambda = 1.34 \times 10^{-122}$ (Planck units, Doc 182)
- Hubble constant $H_0 = 66.2$ km/s/Mpc (Doc 182)
- Geometric path-lengthening redshift mechanism (FEM simulation, Doc 026)

What FFGFT does not formulate

FFGFT does not have an explicit *selection principle* of the RSG type. It derives the scale structure of what exists, but it does not formulate why those structures persist as observable histories rather than being absorbed into the vacuum. The persistence question — why lepton-type histories survive and not others — is answered implicitly through the winding structure of T^4, but not through a dedicated survival-filter formalism.

.3 The Structural Comparison

Dimension	RSG (Austin)	FFGFT (Pascher)
RA4 Primitive	History + survival filter. No committed field content or Lagrangian.	T^4 winding geometry. Committed Lagrangian: $\mathscr{L}_{\text{T0}}$.
RA3 Admissibility	Histories must survive recursive application of the filter above threshold. General condition, no fixed coupling constants.	Only winding configurations consistent with ξ are admissible. $D_f = 3 - \xi$, Z3^3 constraint fixes three generations.
RA2 Representation	Weighted history space with survival measure S_i. No fixed Hilbert space.	$L^2(T^4)$; Fourier modes with winding numbers as natural basis; QM operator algebra derived as projection (Doc 230/231).

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Dimension	RSG (Austin)	FFGFT (Pascher)
RA1.5 Operationalization	Γ_i, W_i as framework parameters — not derived from a fixed input. Scale of the filter undefined without additional input.	All SM constants from ξ alone: masses, α, T_{CMB}, Λ, H_0. No fitting.
RA1 Predictions	Structural persistence conditions. No specific numerical predictions for particle masses or cosmological constants.	Concrete numerical values for all SM constants and cosmological parameters. Explicit falsification criteria.
Redshift	Not addressed explicitly in RSG core.	Geometric path-lengthening in ξ-field, FEM simulation confirms frequency independence. Indistinguishable from metric expansion from inside (Doc 026).
Selection principle	Explicit: survival filter $\Gamma_i W_i$ defines which histories persist.	Implicit: winding number admissibility. No dedicated persistence operator formulated separately from the Lagrangian.
Scale derivation	Not present: Γ_i, W_i require external input for specific values.	Complete: ξ from three mass ratios; all scales follow without additional input.

.4 Is RSG Explainable by FFGFT?

Partially. FFGFT provides the scale structure that RSG requires as input. If $\Gamma_i = \xi \cdot m_i$ (coupling proportional to mass, as in the $\mathscr{L}_{\text{T0}}$ vertex), then RSG's filter parameters are determined by FFGFT's coupling structure. In that reading, RSG's survival filter is the phenomenological description of what FFGFT's Lagrangian produces at the level of observable history selection.

What FFGFT does not supply is RSG's general formulation of the selection principle as a framework-independent operator. RSG's survival filter applies to any history regardless of whether the underlying dynamics comes from FFGFT, from standard QFT, or from some other field theory. That generality is RSG's contribution; FFGFT's specific coupling structure is one instantiation of it.

.5 Is FFGFT Explainable by RSG?

No. RSG does not derive numerical values for particle masses, coupling constants, or cosmological parameters. The electron mass $m_e = 0.511$ MeV, the ratio $m_p/m_e = 1836.15$, the fine structure constant $\alpha \approx 1/137$ — none of these follow from RSG's survival filter without specifying Γ_i and W_i externally. RSG's formalism is consistent with any value of these constants; it does not predict them.

FFGFT, by contrast, fixes ξ from three measured ratios and derives all others. This is a stronger claim, not a weaker one. FFGFT is not a computational overlay of RSG — it is a competing and more constrained answer to the question of what the physical constants are and why.

.6 Which Framework Is More Complete?

The question of completeness depends on what "complete" means.

If completeness means *numerical predictive power* — the ability to derive specific values for observables from minimal input — then FFGFT is more complete. It produces quantitative predictions that RSG cannot make without external specification of its parameters.

If completeness means *generality of the selection principle* — the ability to characterise which structures persist across different underlying dynamics — then RSG is broader. Its survival filter applies regardless of the specific field theory; FFGFT's winding structure is one particular realisation of a persistent configuration, not a general account of persistence.

The two frameworks are therefore asymmetrically complementary:

- FFGFT provides the *scale structure* that RSG's filter requires as input but does not derive.
- RSG provides the *selection-principle language* that FFGFT instantiates but does not formulate as a general operator.

Neither framework is an overlay or bridge layer of the other. They are two distinct frameworks that address adjacent but non-identical questions, with partial structural overlap at the level of persistence conditions.

.7 The Asymmetry in One Sentence

Structural Asymmetry

RSG asks: *which histories survive?* — without specifying the scale. FFGFT asks: *what are the scales?* — without formulating a general persistence operator. RSG needs FFGFT's scales to be quantitative; FFGFT needs RSG's language to explain why winding configurations are selected. Neither reduces to the other.

.8 On Visualisation: Two Parallel Entries

The structural analysis in this document yields a clear requirement for any comparative visualisation of the two frameworks.

FFGFT is not a computational overlay of RSG and not a bridge layer within RSG's architecture. It is a standalone framework with its own RA4 primitive (T^4 torus), its own Lagrangian ($\mathscr{L}_{\mathrm{T0}}$), its own parameter ($\xi$), and its own independent numerical predictions that RSG does not supply.

A structurally correct visualisation therefore shows:

- **Two parallel entries at the same structural level** — RSG and FFGFT as independent frameworks side by side, not in a hierarchy.
- **A ξ-path as its own track** — the ξ parameter and the scales derived from it form an independent line in the visualisation, separate from RSG's filter parameters Γ_i and W_i.

- **Correspondence notes at the contact points** — where FFGFT's $\xi \cdot m_i$ determines the value of RSG's Γ_i, and where RSG's persistence language names what FFGFT's winding structure implicitly selects.

This architecture — two co-equal entries with explicit correspondence arrows — is the only representation that does justice to both frameworks: RSG's generality of the selection principle and FFGFT's completeness of scale derivation remain both visible, without either being subordinated to the other.

.9 Symbol List

Symbol	Meaning
S_i	RSG survival weight of history i
Γ_i	RSG loss rate for history i
W_i	RSG survival weight factor
$\xi = 4/30000$	FFGFT dimensionless coupling constant; fixed by $\kappa = 7$
T^4	4-dimensional torus; FFGFT ontological primitive
$\kappa = 7$	Unique winding exponent from e-p-μ mass triangle
$\mathscr{L}_{\text{T0}}$	T0 Lagrangian (5 terms; Doc 180)
$\Delta m(x,t)$	Dynamical mass field in T0
$D_f = 3 - \xi$	Fractal dimension of physical space (≈ 2.99987)
$L^2(T^4)$	FFGFT Hilbert space
α	Fine structure constant; derived from ξ (Doc 009)
Λ	Cosmological constant from ξ: 1.34×10^{-122} (Doc 182)
H_0	Hubble constant from ξ: 66.2 km/s/Mpc (Doc 182)
RA4–RA1	Layers of Representational Architecture 2.1 (Guevara Calderón 2026)
RSG	Recursive Survival Geometry (Austin 2024–2026)
FFGFT	Fundamental Fractal Geometric Field Theory (Pascher 2024–2026)

Key references: Doc 009 (ξ, $\kappa = 7$) · Doc 025 (CMB) · Doc 026 (redshift, FEM) · Doc 180 (Lagrangian, $\kappa = 7$) · Doc 182 (Λ, H_0) · Doc 230/231 (Hilbert space bridge) · Doc 245 (FFGFT RA diagnostic) · GitHub · Zenodo v1.1.0